

# Chapter 9

## Friction

Friction is the force which resists relative motion between surfaces in contact with each other.

Friction is categorized by the nature of the surfaces in contact and the conditions under which they are interacting. There are many different types of friction, some of which are listed below.

1. **Dry friction**, which is the force that opposes one solid surface sliding across another solid surface.
2. **Rolling friction** is the force that opposes motion of a rolling wheel or ball.
3. **Fluid friction** is the friction between layers of a viscous fluid in motion.
4. **Skin friction**, also called drag, is the friction that occurs between a fluid and a moving surface.
5. **Internal friction** is the force resisting the internal deformation of a solid material.

An understanding of friction is important factor in the design, functionality and performance of many mechanical systems. In this text we will only consider dry friction out of the many types of friction listed above. So far, we have only looked at situations where friction could be safely ignored; however, there are many cases where the effect of dry friction is significant. These include slipping or tipping, blocks, wedges, screws, belts, bearings, and rotating discs. These are the subject of this chapter.

## 9.1 Dry Friction

### Key Questions

- Which types of friction do we study in statics, and which are studied elsewhere?
- What is a *normal* force?
- What is *impending motion*?
- How do you decide when you can use the equation  $F = \mu_s N$  and when you can not?
- Can you show graphically how friction and normal force vectors are related to the friction resultant vector and the friction angle at impending motion?
- What is the friction angle  $\phi_s$ ?
- Why is a distributed normal force represented as a point force that moves as the normal force shifts?

Dry friction, also called **Coulomb friction**, is a force which appears between two solid surfaces in contact. This force is distributed over the contact area and always acts in whichever direction opposes relative motion between the surfaces. We will usually simplify the distributed frictional force by representing it as a single concentrated force acting at a point, as we did in [Section 7.8](#).

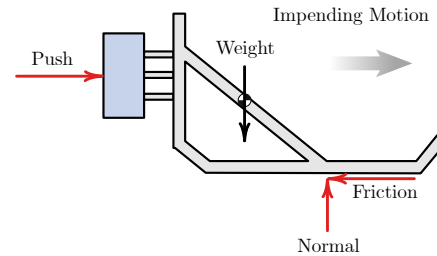
Depending on the details of the situation, dry friction will either hold the object in equilibrium, in which case it is called **static friction**, or it will retard but not prevent motion, in which case it is called **kinetic friction**.

Consider the football training sled shown in [Figure 9.1.1](#). Initially, the sled's weight is supported by a normal force acting on the bottom surface that can be considered as a point force directly beneath the center of gravity; there is no friction force.

When players begin to push the sled, a friction force will appear along the bottom surface which opposes sliding to the right. Both the friction and normal forces can be represented by concentrated horizontal and vertical forces located to the right of the center of gravity. This offset is required to maintain rotational equilibrium against the pushing force. If the players push hard enough, equilibrium will break and the sled will begin to slide in the direction of the push. At this point static friction has transitioned to kinetic friction.



(a) Training sled



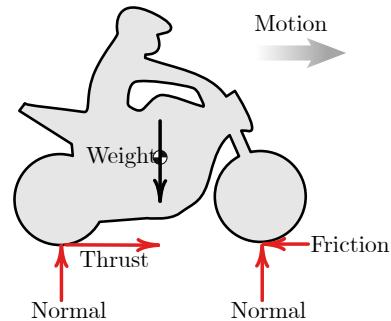
(b) Free-body diagram

**Figure 9.1.1** Static friction holds an object stationary, up to the point of impending motion. Kinetic friction applies to sliding objects.

The rolling friction acting on the motorcycle in [Figure 9.1.2](#) is more complicated. Both wheels rotate clockwise, but the rear wheel is driven by the engine and chain, while the front wheel is rotated by the road friction. The friction force on the rear tire acts to the right and is what propels the bike forward. The dry friction on the front tire acts to the left and retards the motion of the motorcycle.



(a) A motorcycle



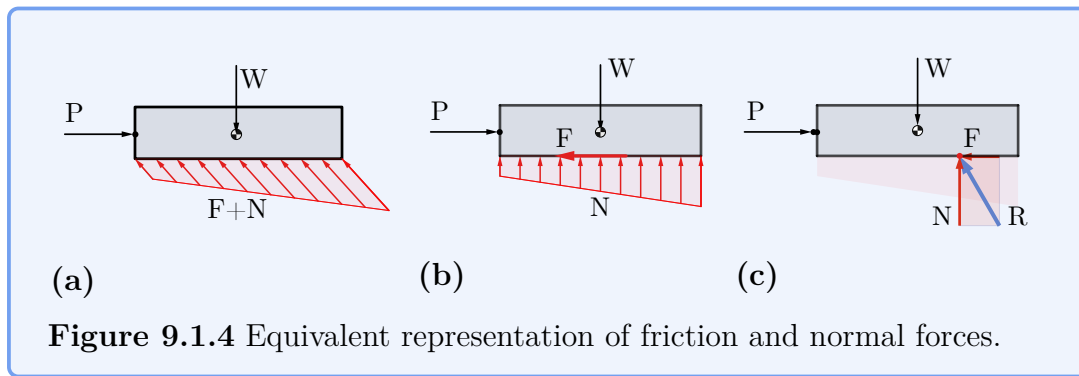
(b) Free-body diagram

**Figure 9.1.2** Rolling friction occurs where the tires contact the road.

### Thinking Deeper 9.1.3 Statically Equivalent Loadings.

The force distribution on the bottom of an object being pushed across a surface is complex, and looks approximately like (a).

To simplify things, we first decompose the actual force into a distributed *normal* force perpendicular to the surface, and a distributed *friction* force parallel to the surface, as shown in the (b). These two distributions are further simplified into two concentrated forces representing the normal and friction components of a single resultant force, as shown in (c). The net resultant force acts at the point required for equilibrium.



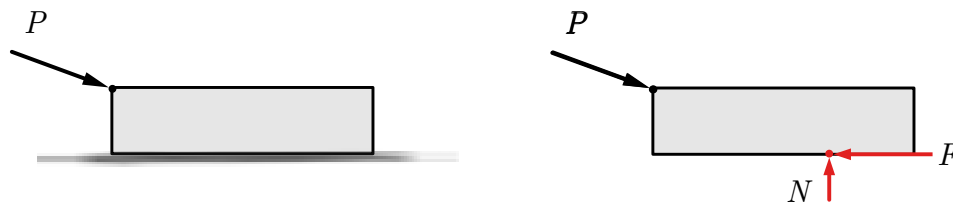
**Figure 9.1.4** Equivalent representation of friction and normal forces.

### 9.1.1 Coulomb Friction

The Coulomb friction model proposes that the force of friction is proportional to the normal force, where the normal force is the force acting perpendicular to the contacting surface.

$$F_f = \mu N$$

The constant of proportionality,  $\mu$ , is called the friction coefficient.  $\mu$  is always greater than zero and commonly less than one. The friction coefficient can be greater than one for materials that exhibit positive adhesion to each other like silicone rubber, glued surfaces, or gecko's feet.

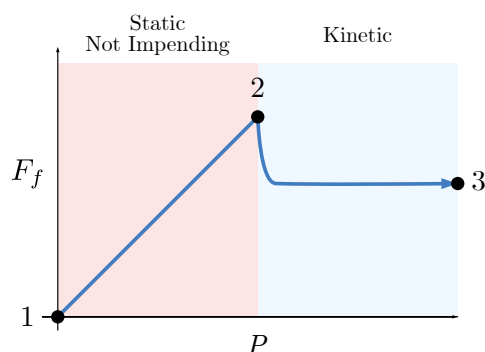


**(a)** Block on rough surface.

**(b)** FBD showing normal and friction forces.

**Figure 9.1.5**

Friction has two distinct regions as shown in Figure 9.1.6, and the value of  $\mu$  is different in each region. The region from point one to point two, where the force of friction increases linearly with load is called the **static friction** region. Here you must use the **coefficient of static friction**  $\mu_s$ . The region from point two to point three, where the friction remains roughly constant is called the **kinetic friction** region. In this region you must use the **coefficient of kinetic friction**  $\mu_k$ .



**Figure 9.1.6** Phases of Coulomb friction.

The coefficient of friction suddenly drops at point two, causing the friction force  $F_f$  to drop as well. Point two is called the point of **impending motion**, because here the situation is unstable. If the applied force increases ever so slightly, the opposing friction force suddenly decreases and the object begins to move.

To better understand the behavior of Coulomb friction imagine an object resting on a rough surface as shown in Figure 9.1.5(b). When force  $P$  is gradually increased from zero, the normal force  $N$  and the frictional force  $F_f$  both change in response. Initially both  $P$  and  $F_f$  are zero and the object is in equilibrium. The interaction between the two surfaces in contact means that friction is available but it is not engaged  $F_f = P = 0$ .

As  $P$  increases, the opposing friction force  $F_f$  increases as well to match and hold the object in equilibrium. In this static-but-not-impending phase  $F_f = P$ .

When  $P$  reaches point two, motion is impending because friction has reached its maximum value.  $F_{f_{\max}} = \mu_s N = P$ . If force  $P$  increases slightly beyond  $F_{f_{\max}}$ , the friction force suddenly drops to the kinetic value  $F_f = \mu_k N$ . The applied force exceeds the frictional force breaking equilibrium and causing the object to accelerate, and accelerating bodies are beyond the scope of Statics!

Notice that static friction at impending motion is greater than kinetic friction, because the coefficient  $\mu_s > \mu_k$  for most materials. Practically, this tells us that once a material starts to move it is easier to keep moving than it was to get it started from rest.

If you wonder why we include kinetic friction in a statics course, remember that a sliding body moving at constant velocity is in equilibrium.

### 9.1.2 Friction Angle and Friction Resultant

Recall that a resultant is the sum of two or more vectors. The **friction resultant** is the vector sum of the friction and normal forces. Since these two forces are perpendicular, the magnitude of the friction resultant can be found using the Pythagorean theorem.

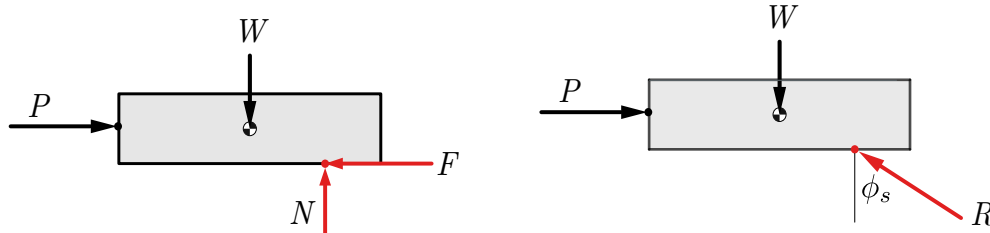
$$R = \sqrt{F_f^2 + N^2} \quad (9.1.1)$$

The **friction angle**  $\phi_s$  is defined as the angle between the friction resultant and the normal force. At impending motion, the friction angle reaches its maximum value. The friction resultant and friction angle are used for screw, flexible belt, and journal bearing type problems.

The maximum friction angle  $\phi_s$  is directly related to the coefficient of static friction  $\mu_s$  since the friction angle  $\phi_s$  is the internal angle of the right triangle formed by the normal force  $N$ , the friction resultant  $R$ , and the friction force  $F$ . Hence:

$$\tan \phi_s = \frac{F}{N} \quad \phi_s = \tan^{-1} \left( \frac{F}{N} \right) \quad (9.1.2)$$

In [Figure 9.1.7](#) a block of weight  $W$  is pushed sideways by force  $P$ . The reaction forces can be represented as separate friction and normal forces, or as combined friction force  $R$  acting at friction angle  $\phi_s$ , measured from the normal direction.



(a) Force components.

(b) Resultant and Friction angle.

**Figure 9.1.7**

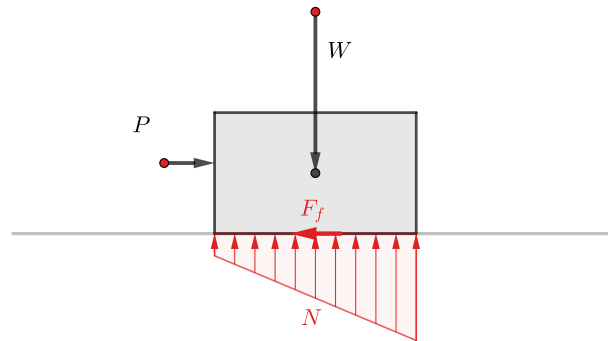
### 9.1.3 Normal Forces

The normal force supporting the object is distributed over the entire contact surface, however it is common on two dimensional problems to replace the distributed force with an equivalent concentrated force acting at a particular spot on the contacting surface, as we did in [Chapter 7](#). This point is rarely at the exact center of the contact surface.

This is illustrated in [Figure 9.1.8](#) where we see in that an object at rest has a uniformly distributed normal force along the bottom surface and the resultant normal force is located directly below the weight. The friction force is not engaged. When a pushing force  $F_{\text{push}}$  is applied, the distributed normal force changes shape, and the resultant normal force  $N$  shifts to the right to maintain equilibrium. The resultant normal force continues to shift to the right the harder you push.

This can be understood from the principle of a two-force body. As we discussed [previously](#), the combined push and weight can be represented a single load acting down and to the right. The friction and normal forces can be combined into a single reaction force acting up and to the left at a point on the bottom surface. These two forces must share the same line of action to maintain equilibrium, so as the pushing force increases, the friction angle changes and

the point shifts to the right. If the point shifts off the physical object then the required friction is greater than the friction available and motion begins.



[Live Link](#)

**Figure 9.1.8** Distributed Normal force changes with load and weight.

### 9.1.4 Coulomb Friction Examples.

**Static, but not impending motion.** For these problems  $F_f$  is independent of  $N$  so you will need one additional piece of data in order to solve for both  $F_f$  and  $N$  since you will not be able to apply the equation  $F_f = \mu_s N$ . The problem statement will use words like ‘sitting’, ‘static’ or ‘at rest’, but no extremal language like ‘maximum’. The reaction can be modeled as a pin reaction, and then possibly checked to confirm  $F < \mu_s N$ .

#### Example 9.1.9

A moment of  $20 \text{ N}\cdot\text{m}$  is applied to a wheel held static by a brake arm. What is the friction force between the wheel and the brake arm?

#### Example 9.1.10

A box sits on a slope, find the resultant of the friction and normal forces on the box.

**Impending motion.** In this case, the friction force will be given by  $F_f = \mu_s N$ . The problem statement will mention maximum or minimum values.

## Example 9.1.11

What is the maximum force applied to the box before it will start to move?

## Example 9.1.12

What is the minimum coefficient of static friction that will keep the box static? What is the lightest box which will not slip or tip on this slope?

## Example 9.1.13

What is the lightest box which will not slip or tip on this slope?

**Kinetic friction.** In these problems,  $F_f = \mu_k N$ . The problem statement will say that the body is moving at a constant velocity.

## Example 9.1.14

A 40 kg box is sliding down a  $20^\circ$  slope, what is the coefficient of friction to keep the velocity constant?

## Example 9.1.15

A rope slips over a surface at constant velocity, what is the contact angle of the rope?

## 9.2 Slipping vs. Tipping

## Key Questions

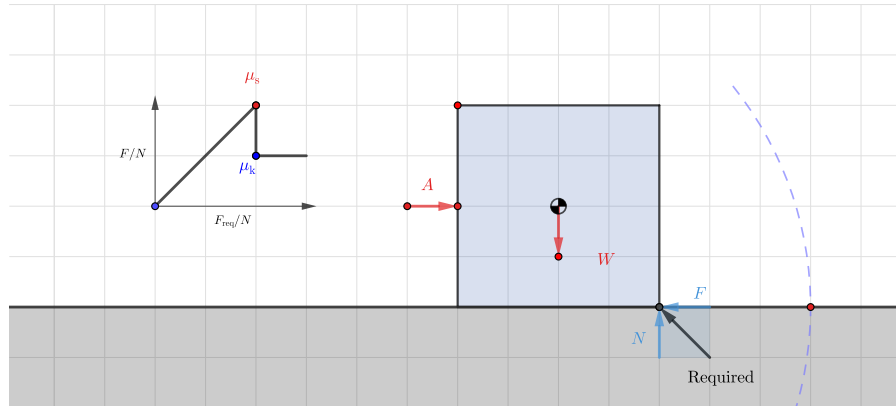
- What are the possible motions which can occur when you apply a force to an object sitting on a rough surface?
- How do you determine which motion will occur in a particular situation?
- Which types of motion are beyond the scope of Statics?

This section focuses on the various ways a rigid body in equilibrium might begin to move. The point at which an object starts to move is called the point of impending motion.

The interactive in [Figure 9.2.1](#) shows a box sitting on a rough surface. Imagine that we start pushing on the side of the box with a gradually increasing

force. Initially, friction between the block and the incline will increase to maintain equilibrium, and the box will sit still.

As we continue to increase the force there are two possibilities; the maximum static friction force will be reached and the box will begin to slide, or the pushing force and the friction force will create a sufficient couple to cause the box tip on its corner.



[Live Link](#)

**Figure 9.2.1** Slipping vs. Tipping

The easiest way to determine whether the box will slip, tip, or stay put is to solve for the maximum load  $P$  twice, once assuming slipping and a second time expecting tipping, then compare the actual load to these maximums. This process is summarized in the following three steps:

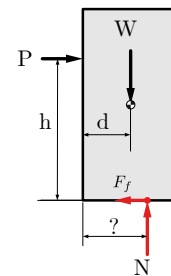
1. *Check slipping.*

As in all dry friction problems, the maximum friction force is equal to the static coefficient of friction times the normal force

$$F_{f_{\max}} = \mu_s N.$$

Assume that the maximum normal force  $N$  is acting at an unknown location and solve for the applied force which will maintain equilibrium. If the load exceeds this value then this the body will slip or maybe tip.

2. *Check tipping.*

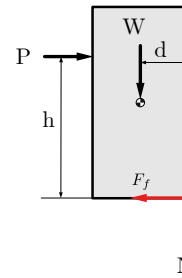


**Figure 9.2.2** FBD to check slipping.

The object will tip when the resultant normal force  $N$  shifts off the end of the object, because it no longer acts on the object so it can't contribute to equilibrium.

Create a free-body diagram assuming that the normal force  $N$  acts at the far corner of the box and solve for the applied force which will maintain equilibrium. Any greater force will make the body tip, unless it is already slipping.

At tipping, the friction force is static-but-not-impending as it has not reached impending motion for slipping.



**Figure 9.2.3** FBD to check tipping.

### 3. Compare the results.

If  $P$  exceeds the smaller of the limiting values, it will initiate the corresponding impending motion.

#### Thinking Deeper 9.2.4 Failure in Engineering.

The goal of engineering design is to forecast and plan for all the ways that something can fail. The challenge is to know the questions to ask and the data to gather to model all possible failure modes. The controlling failure is the mode which occurs at the smallest load.

## 9.3 Wedges

### Key Questions

- Why is the normal force always perpendicular to the contact surface while the friction force always lies parallel to it?
- Can you demonstrate graphically that components of friction and normal forces are related to the right-triangle trigonometry terms sine and cosine?

A wedge is a tapered object which converts a small input force into a large output force using the principle of an inclined plane. Wedges are used to separate, split or cut objects, lift weights, or fix objects in place. The mechanical advantage of a wedge is determined by the angle of its taper; narrow tapers have a larger mechanical advantage.

Wedges are used in two primary ways:

*Low friction wedges* are a simple machines which allows users to create large output forces to move objects using comparatively small input forces. In the log splitter in [Figure 9.3.1\(a\)](#), hydraulic ram pushes a log into a stationary wedge.

The normal force pushes the two halves of the log apart while the friction force  $F_f$  opposes the pushing force  $P$ .

*High-friction (self-locking) wedges* control the location of objects or hold them in place. Examples include doorstop wedges and carpentry wedges. The sailor in [Figure 9.3.1\(b\)](#) is hammering two wooden wedges towards each other to create large compressive forces to secure shoring timbers during a damage control operation.



(a) A low friction wedge is used to split logs.



(b) High friction wedges are used to secure shoring timbers.

**Figure 9.3.1** Wedges in use.

Luckily the analysis of low- and high-friction wedges are identical and they are quite similar to the multi-force body equilibrium problems we saw in [Chapter 5](#) and [Chapter 6](#). The main difference is the inclusion of friction from all non-smooth contact surfaces. The directions of both the normal and friction forces on the free-body diagrams are defined below.

*Normal forces* act between bodies act perpendicular to the contacting surfaces. All normal forces on a free-body diagram should be pointing towards the body because wedges are always subjected to compression.

*Friction forces* are between bodies which act parallel or tangential to the contacting surfaces and are created by the microscopic or large scale roughness of the surfaces. All friction forces on a free-body diagram should be drawn pointing in the direction which resists relative motion at the point of contact.

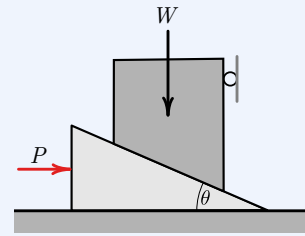
The key added challenge of solving wedge problems is that the angled faces of wedges usually need to be resolved into components in the  $x$  and  $y$ , unless a different coordinate system is used.

One of the critical steps in solving block or wedge problems is to determine which force is engaging the friction of the system. Start by drawing the friction forces on the body where this force acts. As you pass the friction and normal forces to adjacent free-body diagrams, you must always show them as equal and opposite, action-reaction pairs. This is illustrated in the following example.

## Example 9.3.2

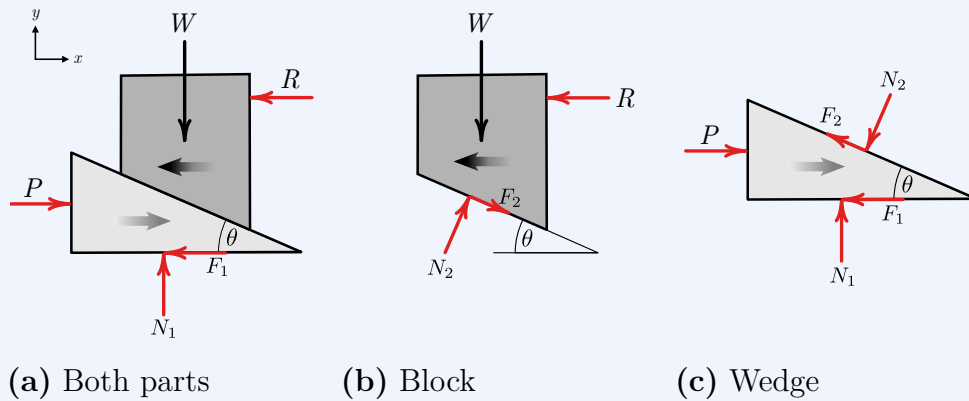
Find the minimum force  $P$  required to start raising the 10 lb block using a  $25^\circ$  wedge.

Assume that the coefficient of static friction at both contact surfaces is  $\mu_s = 0.20$  and that the wedge is massless.



**Solution.** Given:  $W = 10$  lb,  $\mu_s = 0.2$ , and  $\theta = 25^\circ$ .

Start by assuming the wedge is on the verge of moving to the right and drawing free-body diagrams of the system and each of its parts.



**Figure 9.3.3**

By considering equilibrium of the combined wedge and block in the  $y$  direction, we find that

$$N_1 = W = 10 \text{ lb}$$

Assuming that the wedge is pushed to point of impending motion,

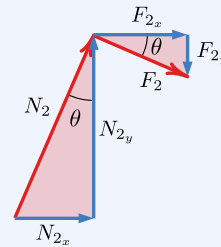
$$F_1 = N_1 \mu_s = 2 \text{ lb.}$$

There are still two unused equilibrium equations for this free-body diagram, but they are not sufficient to find unknown forces  $P$  and  $R$  because the point where resultant friction force acts is also unknown.

Next, consider equilibrium of the block, [Figure 9.3.3\(b\)](#). A detail of  $N_2$  and  $F_2$  in [Figure 9.3.4](#) shows how the their  $x$  and  $y$  components are related to the wedge angle  $\theta$ .

Applying equilibrium in the  $y$  direction with  $F_2 = \mu_s N_2$  gives

$$\begin{aligned}\Sigma F_y &= 0 \\ -W + N_{2y} - F_{2y} &= 0 \\ -W + N_2 \cos \theta - \mu_s N_2 \sin \theta &= 0 \\ N_2(\cos \theta - \mu_s \sin \theta) &= W \\ N_2 &= 12.17 \text{ lb}\end{aligned}$$



**Figure 9.3.4**

Since the wedge is at the point of impending motion,

$$\begin{aligned}F_2 &= \mu_s N_2 \\ &= 2.43 \text{ lb}\end{aligned}$$

With the friction and normal forces on both sides of the wedge known, we can now find the required force  $P$  using the free body diagram of the [wedge 9.3.3\(c\)](#).

$$\begin{aligned}\Sigma F_x &= 0 \\ P - F_1 - F_{2x} - N_{2x} &= 0 \\ P &= F_1 + F_2 \cos \theta + N_2 \sin \theta \\ P &= 9.35 \text{ lb}\end{aligned}$$

### Example 9.3.5

Using the same system as in [Example 9.3.2](#), find the minimum force  $P'$  to prevent the wedge from slipping out from under the block. This is similar to the previous example, but the free-body diagrams need to change in the following ways:

- all friction force directions reverse since the impending motions of the block and wedge have reversed, and
- the direction of  $P'$  may have to reverse if the wedge has sufficient friction to stay static when  $P' = 0$ .

Note that for forces between  $P$  and  $P'$ , the system is static and the friction forces are static-but-not-impending.

**Solution.** This problem still requires a free-body diagram for each block and wedge. The only difference between it and the previous example is that the impending motion for the wedge is now to the *left*. Thus, the

friction forces on the top and bottom of the wedge are to the *right* and are both still at impending motion.  
The remainder of the solution is left as an exercise for the reader.

## 9.4 Screw Threads

### Key Questions

- Can you describe how the right-hand rule relates to the motion and rotation of screws?
- What is the thread pitch and friction angle for a screw?
- Contrast the different types of screw motion, with and against applied loads, and match the motion cases to their corresponding equation.
- Why does a screw and a nut move in relatively opposite directions?

A screw thread is uniform shape which spirals around the inside or outside surface of a cylinder or cone. Like wedges, screws are simple machines. They are essentially a ramp or inclined plane wrapped into a helix, and the input to screws is torque rather than linear force. The mechanical advantage of a screw depends on its **lead**, which is the linear distance the screw travels in one revolution.

Screws used to fix objects in place are called **fasteners**, and screws used to move objects are called **power screws** or **lead screws**. In this chapter we will focus on power screws.

A power screw assembly includes a nut with matching internal threads which fits around the screw. There are two ways that a power screw can operate based on the movement of the screw and nut. In a scissors jack, the operator rotates the threaded rod with a crank fitted to the nut on the right, which is not threaded but acts as a thrust bearing. The nut on the left moves along the stationary screw to raise the load. In the C-clamp, the nut is stationary and the screw translates through as it rotates. In either case, a small moment on the screw can cause enormous forces on the nut, with the added benefit of the force being applied at a precise location as controlled by the screw.



(a) A scissors jack uses fixed screw and a moving a nut.

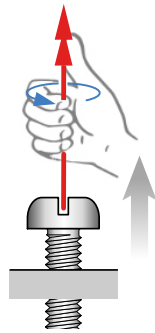


(b) A C-clamp screw rotates and translates through a fixed nut.

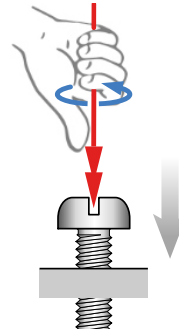
**Figure 9.4.1** Power Screws

### 9.4.1 Screw Motion and the Right-hand rule

Most screw threads are right-hand threaded, which means they follow a right-hand rule as illustrated in [Figure 9.4.2](#). When you use your right hand to then turn a right-handed thread towards your fingertips, it will move in the direction of your thumb. When you are look at the head of a bolt and rotate it clockwise, it tightens i.e. *righty-tighty*. The reverse, *lefty-loosy*, is also true.



(a) Loosening



(b) Tightening

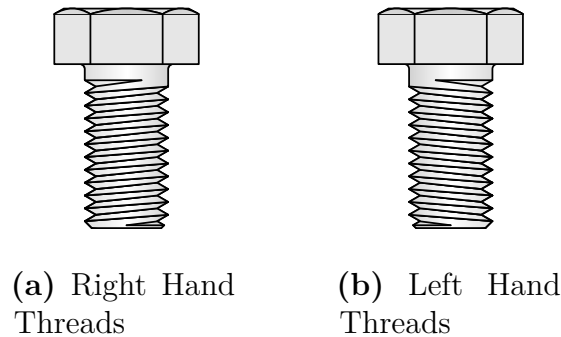
**Figure 9.4.2** Right Hand Threads

Left-handed screws are less common but are found in some applications, especially:

- machines where reverse-threading prevents them from loosening gradually under the torque of the moving part. For example, the left pedal on a bicycle.

- machines where the movement of the screw creates dual motion, like a hand-screw wood clamp.
- situations where you do not want to mix up constituents. Cutting torches use right-hand thread for the oxygen and left-hand threads for the acetylene connections.

The motion of left-handed screws can either be thought of as opposite the right-hand rule or conforming to the same relationships if you use your left hand. Notice that right-handed threads slope up to the right, while left-handed threads slope down. Note that turning a thread upside-down does not reverse its handedness.



**Figure 9.4.3** Screw Thread Handedness

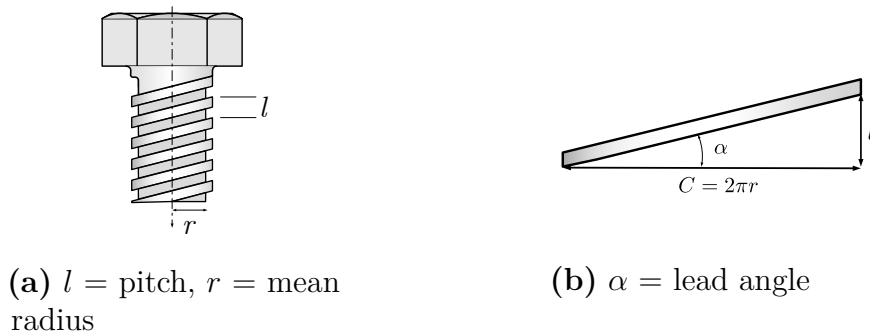
## 9.4.2 Screw Thread Properties

Although most thread profiles are V-shaped, we only consider square threads in this book. General purpose thread profiles, like NPT and ISO metric are more difficult to analyze.

The easiest way to analyze a square-thread power screw system is to turn the problem into a two-dimensional problem by ‘unwrapping’ the ramp from around the cylinder of the shaft. The significant geometrical properties of the thread are

- The **mean radius**,  $r$ .
- The **lead**,  $l$ , also called the **pitch**.
- The **lead angle**,  $\alpha$ , also called the **helix angle**.

To visualize these terms, imagine unwrapping a thread from around the screw, as shown in [Figure 9.4.4](#).



**Figure 9.4.4** Thread properties

The mean radius is the distance from the centerline to a point halfway between the tip and the root of the thread. Twice this value is the effective diameter.

The lead is the linear travel the nut makes in one revolution, which is also the distance from a point on the screw thread to a corresponding point on the same thread after one rotation. Threads are commonly designated by the number of threads per-inch or per-centimeter, and pitch is the inverse of this value.

The lead angle is related to the pitch and the mean radius by trigonometry. Using the right triangle shown in [Figure 9.4.4\(b\)](#), the thread lead angle  $\alpha$  is the inverse tangent of the ratio of the lead over the circumference

$$\alpha = \tan^{-1} \left( \frac{l}{2\pi r} \right)$$

### 9.4.3 Moment to Reach Impending Motion

The focus of this section is to find the magnitude of a moment which will push a screw to impending motion. Impending motion is the threshold between the system holding still and moving, so knowledge of the moment required at impending motion allows you to interpret what happens to the screw system in static but not impending-motion conditions as well.

Assuming that motion is impending means that we can use the coefficient of static friction  $\mu_s$  and the related friction angle  $\phi_s$ . Recall from earlier in this chapter, that the friction angle  $\phi_s$  is related directly to  $\mu_s$  by the equation:

$$\phi_s = \tan^{-1} \left( \frac{F}{N} \right)$$

We now have the tools assembled to derive the relationship between a screw's geometry and the applied loads.

#### Applied Force Opposes Impending Motion

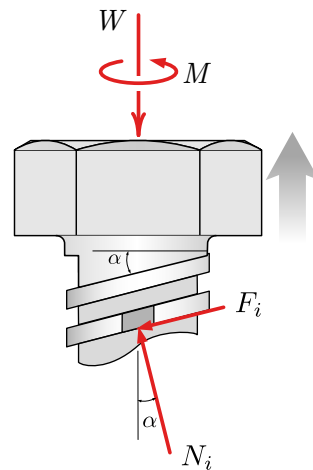
In this scenario we will examine the magnitude of moment  $M$  required to raise a screw to impending motion against the applied force  $W$  as shown in [Fig-](#)

ure 9.4.5(a). To eliminate any references to the orientation of the screw and force like up, down, left, or right, this type of motion will be described as “the applied force opposes impending motion.” This case occurs any time you are applying a force to an object with a screw.

The free-body diagram in Figure 9.4.5(b) shows the moment  $\mathbf{M}$  required to raise a load  $\mathbf{W}$  and the friction and normal forces acting on a slice of thread. These must be summed over the entire length of the thread to find the total friction and normal forces.



(a) Screw Jack



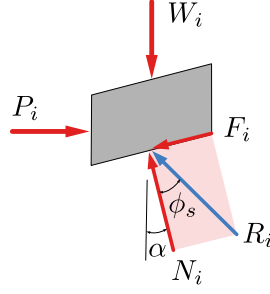
(b) FBD

**Figure 9.4.5** A screw jack supports a load which opposes impending motion.

It is often easiest to express the friction force  $F$  and normal force  $N$  as the friction resultant  $R$  and the friction angle  $\phi_s$ . Recall that the friction force  $F$  direction always opposes the impending motion of the point of contact — in this case, the screw threads. Also, the thread angle  $\alpha$  determines the angle of the normal force  $N$  from the centerline of the screw. Finally, the friction angle  $\phi_s$  is the angle between the friction resultant force  $R$  and the normal force  $N$ .

Figure 9.4.6 is a free-body diagram of single thread element.  $W_i$  is the fraction of the total weight on this element, and the total moment  $M$  is represented by the fraction of the rotational force  $P_i$  acting at the mean radius  $r$  from the center of the screw

$$P_i = M/r.$$

**Figure 9.4.6** FBD of Thread Element

Summing the forces in the  $x$  and  $y$  directions for the free-body diagram in Figure 9.4.6 yields:

$$\begin{aligned} \sum F_x &= 0 & \sum F_y &= 0 \\ P_i &= R \sin(\alpha + \phi_s) & W_i &= R \cos(\alpha + \phi_s) \end{aligned}$$

By summing the forces across all elements of one wrap of the screw we find:

$$\begin{aligned} \sum F_x &= 0 & \sum F_y &= 0 \\ \frac{M}{r} &= \Sigma R \sin(\alpha + \phi_s) & W &= \Sigma R \cos(\alpha + \phi_s) \end{aligned}$$

We next need to reduce these two equations to a single equation and also eliminate the difficulty to quantify  $\Sigma R$  term. Thus we solve both equations for  $\Sigma R$ .

$$\begin{aligned} \Sigma R &= \frac{M}{r \sin(\alpha + \phi_s)} \\ \Sigma R &= \frac{W}{\cos(\alpha + \phi_s)} \end{aligned}$$

Then, set them equal to each other and solve for the moment  $M$ .

$$M = W r \tan(\phi_s + \alpha) \quad (9.4.1)$$

$M$  is the moment required to raise the screw to impending motion,  $W$  is the force load on the screw,  $r$  is the mean radius of the screw,  $\phi_s$  is the screw friction angle, and  $\alpha$  is the screw thread pitch.

Practically, this equation says that the moment to move a screw against an applied force must overcome the screw friction, represented by  $\phi_s$ , and the component of the load on the screw, represented by  $\alpha$ .

### Applied Force Supports Impending Motion

When the impending motion of the screw is in the direction of the applied force, we can also state that the “applied force supports impending motion.” This case occurs any time you remove a force held by a screw, like lowering the load supported by the screw jack in Figure 9.4.5(a).

This situation is a bit more complicated than the previous one, because there are three different possibilities depending on the relative magnitude of the friction angle  $\phi_s$  and the thread angle  $\alpha$ . Cases include

1. *Self-locking.*  $\phi_s > \alpha$ .

In this case the load will not cause the screw to rotate by itself,

2. *Unwind-with-load.*  $\phi_s < \alpha$ .

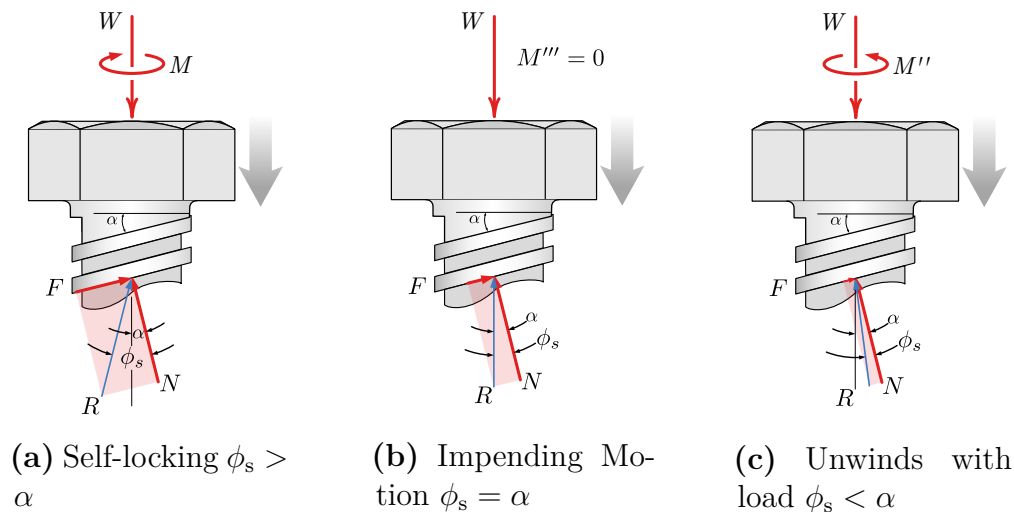
In this case, the load will move the screw without assistance, and

3. *Impending motion.*  $\phi_s = \alpha$ .

This case is the boundary between the two other cases.

In all three cases the thread angle  $\alpha$  is the angle between the normal force  $N$  and the centerline of the screw, and the friction angle  $\phi_s$  is the angle between the friction resultant force  $R$  and the normal force  $N$ .

The derivations of the relationships are quite similar to the derivation of (9.4.1), but use subtly different free-body diagrams for each of the three cases. See Figure 9.4.7 below.



**Figure 9.4.7** Three possible situations when the load acts in the direction of impending motion.

**Self-locking screw.** Self-locking screws are the type of screws that you will encounter most often in mechanical systems as they are highly predictable. They have sufficient friction available to hold their applied load even with no moment

applied. Thus, they can safely carry a load in a static-but-not impending condition until you wish to overcome the excess friction by applying a moment  $M'$  to push them to impending motion.

Summing the forces in the  $x$  and  $y$  directions for the free-body diagram in Figure 9.4.7(a) yields:

$$M' = Wr \tan(\phi_s - \alpha) \quad (9.4.2)$$

**Unwind-with-load screw** As its name implies, an unwind-under-load screw will start turning unless a moment  $M''$  is applied to keep the screw at or beyond impending motion. The moment to push a self-locking screw to impending motion  $M'$  is in the opposite direction as the moment to keep unwind-under-load screws at impending motion  $M''$ , as  $M''$  is in the same direction as the moment to loosen (or raise) a screw. These unwind-with-load screws are not often found in mechanical systems, except for in dynamic motion control systems, where the screw is used to slow down motion.

To be designed in an unwind-with-load condition, a screw must have a quite steep thread angle  $\alpha$  and minimal friction between the threads and nut, which reduces  $\phi_s$ .

Summing the forces in the  $x$  and  $y$  directions for the free-body diagram in Figure 9.4.7(c) yields:

$$M'' = Wr \tan(\alpha - \phi_s) \quad (9.4.3)$$

**Impending-motion screw** As the derived equations for all three unwind-with-load screw cases push the screw towards impending motion, when a screw is already at impending motion, it requires no applied moment to maintain equilibrium; however, this case is mechanically unstable. If the load increases slightly the screw will begin to unwind-under-load, whereas if the load decreases slightly the screw will become self-locking.

Summing the forces in the  $x$  and  $y$  directions for the free-body diagram in Figure 9.4.7(b) yields:

$$M''' = 0 \text{ when } \phi_s = \alpha \quad (9.4.4)$$

The concept of an applied force in the direction of impending motion works equally well for either a force applied in the impending motion direction of a screw, or for a force applied to the impending motion direction of a nut. An example of the first case is the screw jack lowering a load, and the second could be a scissors jack that has a rotating but non-translating screw, plus a non-rotating but translating nut.

## 9.5 Flexible Belts

### Key Questions

- How does the impending motion of the system determine which side of the belt will have a larger tension?
- How do you compute the contact angle  $\beta$  between the belt and pulley or cylinder?
- How do you compute the tension differential on either side of the belt or cylinder for both flat and v-belts?
- What determines the maximum torque transfer available from a pulley system?

When a belt, rope, or cable is wrapped around an object, there is potential for flexible belt friction. In [Figure 9.5.1\(a\)](#) friction allows the sailors to control the speed that the mooring line pays out. The friction between the line and the bollard depends on the number of turns the line takes around the bollard post. In [Figure 9.5.1\(b\)](#) friction forces prevent the belt from slipping allow it to transfer power from the motor to the drive pulley.



(a) Sailors handling a mooring line.

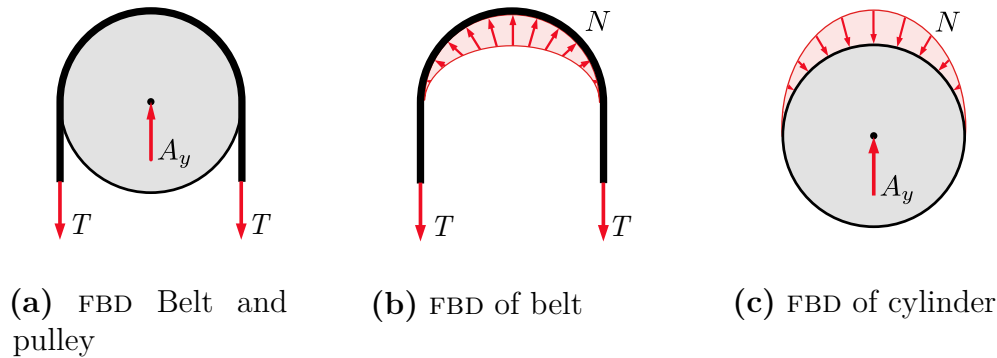


(b) Power transmission pulleys.

**Figure 9.5.1** Applications of belts.

### 9.5.1 Frictionless Belts

Imagine with a flat, massless cable or belt passing over a frictionless cylinder or pulley, in equilibrium. (Figure 9.5.2). A non-uniform distributed normal force acts at points of contact with the cylinder to oppose the tension in the belt and maintain equilibrium. The normal force varies as a function of the angle between the contact surface and the direction the belt tension.



**Figure 9.5.2** Free-body diagrams, equal tension.

Without friction, the two tensions must be equal otherwise the belt would slip around the cylinder. The only interaction force between the belt and pulley is the distributed normal force. Due to the symmetry of this example, the  $x$  components of the distributed force all cancel and the resultant is purely vertical. In other situations this will not be true.

### 9.5.2 Friction in Flat Belts

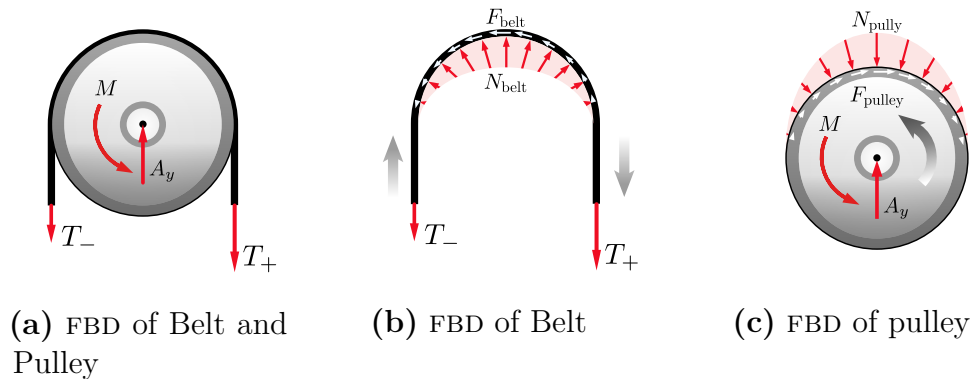
When friction is available to oppose the sliding, the tensions in the two sides of the belt will not be equal and friction will cause the pulley to rotate. Alternately if the pulley is driven by an external moment, friction will cause one tension to increase and the other to decrease. This, of course, is the point of a belt and pulley system — power transmission from the belt to the pulley or vice-versa.

Figure 9.5.3(a) shows a free-body diagram of a belt and pulley in equilibrium. The net moment caused by the two belts and the applied moment  $M$  are in balance. The system may be stationary, or it may be rotating at a constant velocity; however, it is impossible to tell from this diagram which direction.

Figure 9.5.3(b) shows a free-body diagram of the belt. Since  $T_+$  is greater than  $T_-$ , in the absence of friction, the left side will move up, and the right side will move down, as indicated by the arrows which indicate the relative motion of belt with respect to the pulley. A distributed friction force  $F_{\text{belt}}$  between the belt and pulley which opposes the relative motion and maintains rotational equilibrium. A distributed normal force also exists to maintain equilibrium in the  $y$  direction.

Figure 9.5.3(c) shows a free-body diagram of the pulley. The frictional and normal forces acting on the pulley are equal and opposite to those acting on the

belt. The arrow indicates that the impending relative motion of the pulley with respect to the belt is counter-clockwise. The actual direction of rotation is not known or indicated. Friction always acts opposite to the direction of relative motion.



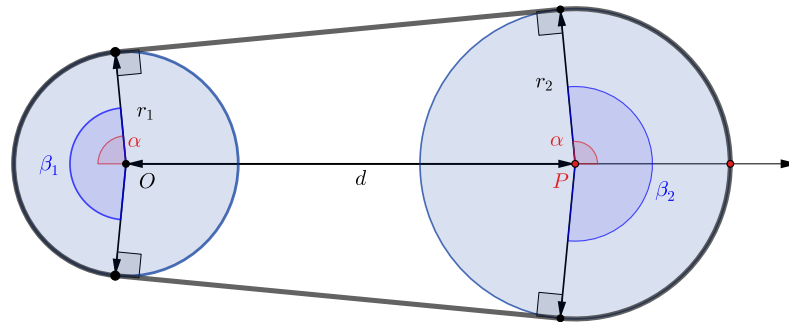
**Figure 9.5.3** Unequal Tensions

Increasing  $M$  or the belt tension ratio  $T_+/T_-$  will increase the power transmitted until the belt starts to slip. This occurs when the friction increases to the maximum available value  $F_{\max} = \mu_s M$ . At this point, motion (slipping) is impending.

We are interested in determining the range of values for the tension forces where the belt does not slip relative to the surface. For a flat belt, the maximum value for  $T_+$  depends on the magnitude of  $T_-$ , the static coefficient of friction between the belt and the surface  $\mu_s$ , and the contact angle  $\beta$  between the belt and the surface.

### Contact Angle $\beta$

You will need to use the geometry of the pulleys to find the contact angle  $\beta$  between the belt and pulley. The belt will depart the pulley at a point of tangency, which is always perpendicular to a radius. To find  $\beta$  create one or more right triangles using the incoming and outgoing belt paths and apply complementary angles to relate the belt geometry to the contact angle. There is no simple rule for transferring cable angles over to the contact angle, but in general, extend radial lines from the center of the pulley out to the belt's tangential lines. Next, create right triangles with each radial line and work to find all the angles which add up to the contact angle  $\beta$ .



[Live Link](#)

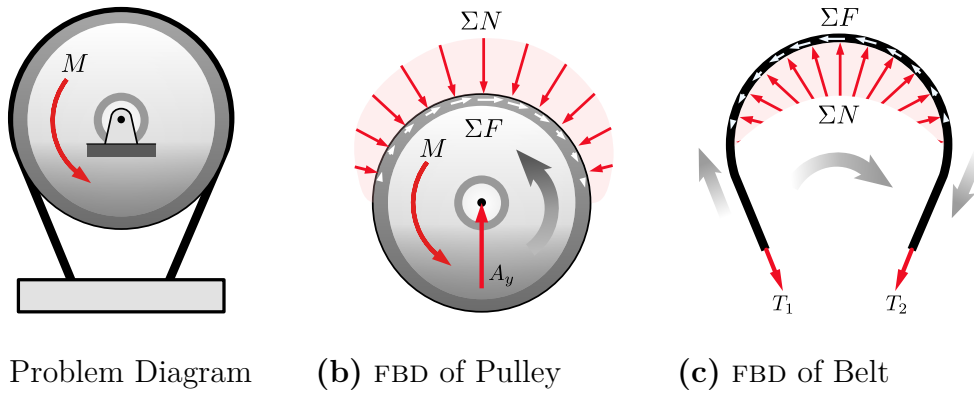
**Figure 9.5.4** Relationship between pulley geometry and contact angle  $\beta$ .

### Belt Tension

We have seen that when there is friction between a belt and a pulley, the tensions on either end of the belt are not the same. In previous problems, we simply guessed a direction for an unknown force, and then used the sign of the numerical answer to confirm or reverse our assumption. Unfortunately, this does not work for flexible belts, where we must make the correct determination before starting computations. So, how can we determine which side has the larger tension, and which side is smaller?

The following discussion guides you through two methods to make this decision. [Figure 9.5.5](#) shows a pulley and belt system and the associated free body diagrams. The pulley is driven by a motor which supplies a counterclockwise moment of  $M$ . The belt is fixed, and holds the pulley in equilibrium until slipping occurs. Both the pulley and belt are assumed massless.

The grey arrows indicate the direction of *impending* motion. This is the motion which will occur if the belt slips. For the belt, which is fixed, this impending motion is *relative* to the pulley. For the pulley, motion is impending *relative* to the belt, but since it can actually rotate it also has *absolute* impending motion with respect to the earth.



**Figure 9.5.5** Motor drive pulley with a fixed belt.

1. *Method 1: Draw free-body diagrams and sum tensions along the cable.*

Friction always opposes impending relative motion at the point of contact, so if you can determine the direction the belt will potentially slip, you also know the direction of the friction force. You can find out which tension is larger and which is smaller by drawing a free-body diagram of the belt and summing forces along it.

The free-body diagram [Figure 9.5.5\(b\)](#) shows the forces acting on the pulley, which are:

- Reaction force  $A_y$  from the fixed center axle.
- A distributed normal force  $\Sigma N$  acting radially along the contact surface with the belt. <sup>1</sup>
- A distributed friction force  $\Sigma F$  acting along the contact surface, opposing moment  $M$  and the impending motion of the pulley.

The free-body diagram [Figure 9.5.5\(c\)](#) shows the forces acting on the belt, which are:

- The belt's internal tension forces, labeled  $T_1$  and  $T_2$  since at this point we don't know their relative magnitudes.
- A distributed normal force  $\Sigma N$  acting radially along the contact surface between the pulley and belt. These are the distributed normal forces on the pulley transferred equal-and-opposite to the belt.
- A distributed friction force  $\Sigma F$  acting along the surface of the belt, again equal-and-opposite to the corresponding forces on the pulley. Since the belt is not actually moving these forces oppose the belt's *relative impending motion* with respect to the pulley.

<sup>1</sup>The distributed normal forces is not symmetrical as drawn, but actually biased towards the right, in order to have the net leftward force required to oppose the net friction acting to the right. Fortunately, the actual shape is not significant to us.

Summing forces along the belt, we find that the tension  $T_1$  plus the distributed friction force  $\Sigma F$  must equal  $T_2$  for equilibrium.

$$\begin{aligned}\Sigma F_{\text{belt}} &= 0 \\ T_1 + \Sigma F - T_2 &= 0 \\ T_2 &= T_1 + \Sigma F\end{aligned}$$

Therefore, the larger tension  $T_+ = T_2$  and the smaller tension  $T_- = T_1$ .

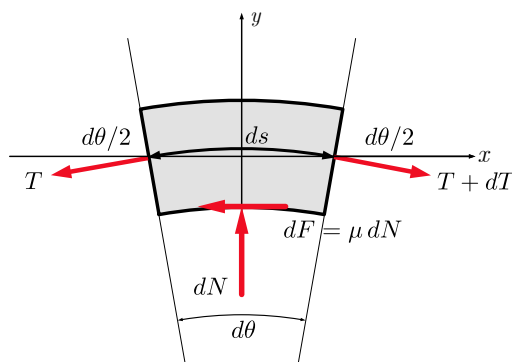
2. *Method 2: Larger tension acts in the direction of the impending motion of the belt.*

Following the logic of Method 1, it turns out that the larger tension always points in the direction of impending motion of the belt. It does not matter if the impending motion is relative as here with a fixed belt, or absolute as when the belt moves around a fixed object.

There are multiple ways to determine the smaller and larger tensions in a flexible belt system. You can use the interactive below to develop your intuition on the relationship between belt tension, pulley moment, friction and relative motion.

### Change in Belt Tension due to Friction

Applying the equilibrium equations to a free-body diagram of a differential element of the belt enables us to derive the relation between the two belt tensions, the contact angle  $\beta$ , and the friction coefficient  $\mu_s$ .



**Figure 9.5.6** FBD of a differential element of a flexible belt.

Summing the forces in the  $x$  direction gives

$$\begin{aligned}\sum F_x &= 0 \\ \mu dN + T \cos(d\theta/2) &= (T + dT) \cos(d\theta/2) = 0 \\ \mu dN &= dT \cos(d\theta/2)\end{aligned}$$

As  $d\theta$  approaches zero,  $\cos(d\theta/2)$  approaches one, so in the limit,

$$dN = \frac{dT}{\mu}$$

And summing forces in the  $y$  direction gives

$$\begin{aligned}\sum F_y &= 0 \\ dN &= T \sin(d\theta/2) + (T + dT) \sin(d\theta/2) \\ dN &= (2T + dT) \sin(d\theta/2) \\ &\approx 2T(d\theta/2) + dT(d\theta/2)\end{aligned}$$

where we have used the small angle approximation  $\sin(d\theta/2) \approx d\theta/2$ . Dropping the second order differential term  $dT d\theta$  as negligible, yields

$$dN = T d\theta.$$

Solving simultaneously by eliminating  $dN$  leaves us with

$$\frac{dT}{T} = \mu d\theta,$$

which we can integrate between  $T_-$  and  $T_+$  to find

$$\begin{aligned}\int_{T_-}^{T_+} \frac{dT}{T} &= \mu \int_0^\beta d\theta \\ \ln \frac{T_+}{T_-} &= \mu\beta\end{aligned}$$

Integrating both sides gives:

$$\frac{T_+}{T_-} = e^{\mu\beta}, \quad (9.5.1)$$

where  $e$  is the natural log base 2.718,  $\mu$  is the friction coefficient between the belt and pulley, and  $\beta$  is the contact angle between the belt and pulley in radians. The larger this ratio is, the more torque the belt can transmit.

Notice that the belt tension ratio is independent of the surface size and shape, provided the belt makes continuous contact.

#### Example 9.5.7 Rope Friction.

How much force is necessary to support a 100 lb weight suspended by a rope wrapped four and half times around a horizontal bar? Assume the coefficient of friction  $\mu \approx 0.23$ .

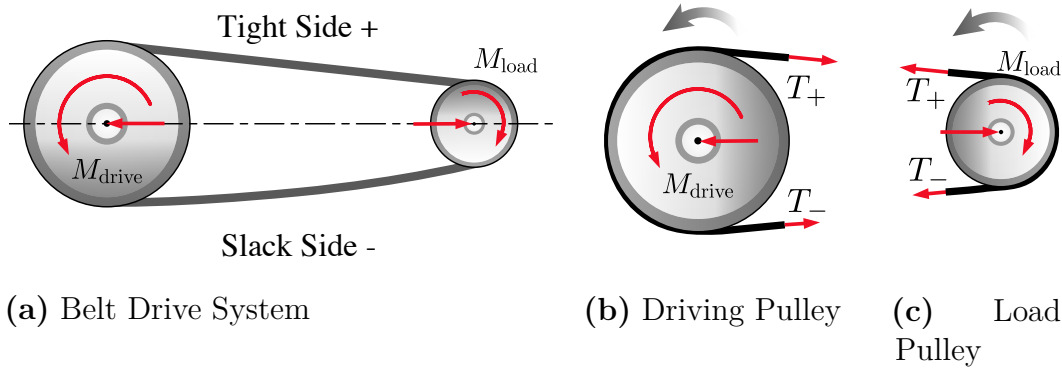


**Solution.** Apply (9.5.1) with  $\mu = 0.23$  and  $\beta = 4.5 \times (2\pi) = 28.27$  radians.

$$\begin{aligned}\frac{T_+}{T_-} &= e^{\mu\beta} \\ T_- &= \frac{100 \text{ lb}}{e^{0.23 \times 28.27}} \\ &= 0.15 \text{ lb}\end{aligned}$$

### 9.5.3 Torque in Belt Systems

A belt-driven system consists of an input pulley driven by a rotational power source and one or more output pulleys driving loads. The maximum torque that can be transmitted by the system is determined by the maximum value for  $T_+$  before slipping occurs at either the input or any output pulley. We will need to consider each of the pulleys independently.

**Figure 9.5.8** Unequal Tensions

Start by solving for the resting tension  $T_-$ . This is the tension the belts prior to any motion or power transfer. Practically, machines provide adjustments to pre-tension the belt to insure sufficient normal force when started. When we turn on the machine and increase the torque, the resting tension remains constant while the tension on the drive side  $T_+$  increases.

If the pulleys have the same coefficients of friction, it can be assumed that the belt will first slip at the smaller of the two pulleys as the smaller pulley has a smaller contact angle  $\beta$ . See [Figure 9.5.4](#).

Once we have the maximum value for  $T_-$ , we can use that to find the maximum input and output moments. Next, to find the torque, we then find the net moment exerted by the two tension forces, where the radius of the pulley is the moment arm.

The maximum input torque  $M_i$  before slipping is

$$M_i = (T_{+\max} - T_-)r_i$$

The maximum output torque  $M_o$  before slipping is

$$M_o = (T_{+\max} - T_-)r_o$$

In a rotating shaft, power is equal to the torque times the angular velocity in radians per second.

$$P = T\omega, \quad (9.5.2)$$

so

$$P_{\max} = M_i(\omega_i) = M_o(\omega_o). \quad (9.5.3)$$

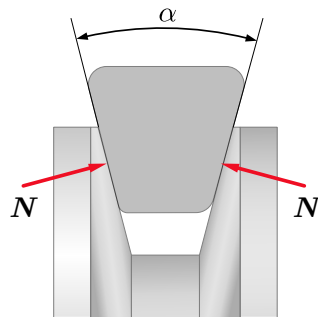
Unlike the torque which steps up or down based on the pulley radii, the input and output powers are equal to each other, ignoring all efficiency losses.

### 9.5.4 V-Belts

A flat belt pulley interacts with one surface of the belt. A V shaped pulley allows the belt to wedge tightly in the groove, increasing friction and torque transmission. A V-belt's enhanced friction comes from the increased normal forces which are a function of the groove angle  $\alpha$ .

The sum of the normal force vertical components is the same for a flat belt or V-belt. However, the horizontal components of the normal forces in a V-belt, effectively pinch the belt, thereby increasing the available friction force. The belt should not contact the bottom of the groove, or else the wedge effect is lost

$$\frac{T_+}{T_-} = e^{\left( \frac{\mu_s \beta}{\sin(\alpha/2)} \right)}. \quad (9.5.4)$$



**Figure 9.5.9** V-belt and pulley cross section.

As we can see from the equation above, smaller groove angle and steeper sides result in a larger maximum tension ratio, resulting in higher torque transmission. The trade-off with steeper sides, however, is that the belt becomes wedged more firmly in the groove and requires more force to unwedge itself as it leaves the pulley. This unwedging force decreases the efficiency of the belt-driven system. An alternate design choice would be a chain-drive which carries very high-tension differences efficiently.

## 9.6 Journal Bearings

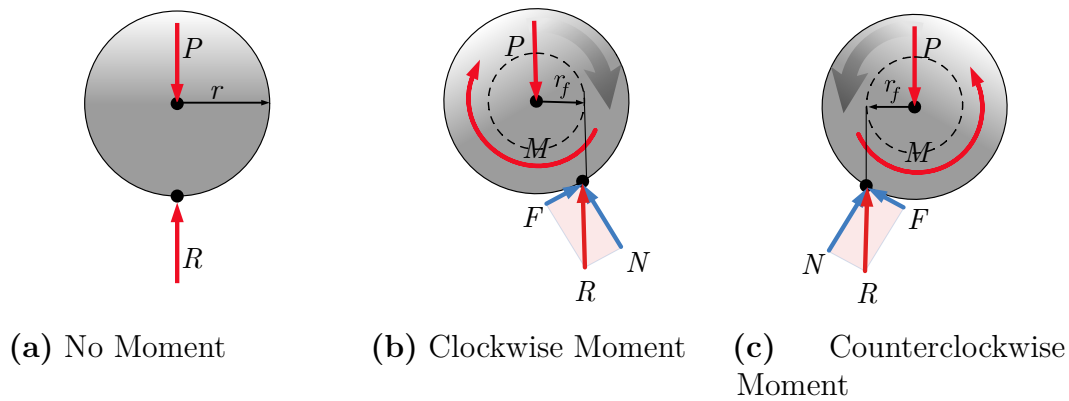
### Key Questions

- Why does the point of contact between a shaft and a journal bearing shift as the shaft rotates?
- Why is the resultant contact force tangent to the friction circle?
- Can you draw appropriate free-body diagrams of journal bearing systems and solve for unknown values?

### 9.6.1 Journal Bearing Friction

A bearing is a machine element used to support a rotating shaft. Bearing friction exists between the rotating shaft and the supporting bearing. Though other types of bearings exist including, ball, roller and hydrodynamic, we will focus on *dry friction* journal bearings. Oil lubricated journal bearings require a knowledge



**Figure 9.6.2** Shaft Free-body Diagrams

The most straightforward process to relate the load, normal and friction forces for a journal bearing is by performing the following steps:

1. Assume that the shaft and bearing opening have the same radius, but draw the shaft a bit smaller to emphasize the contact point at the point of tangency.
2. Combine the normal and friction forces into a single friction resultant force

$$\mathbf{R} = \mathbf{F} + \mathbf{N}.$$

3. Determine the radius of the friction circle,  $r_f$ , which is a circle around the center tangent to the friction resultant  $R$ . The friction circle radius is a function of the shaft radius  $r$  and the friction angle  $\phi_s$ .

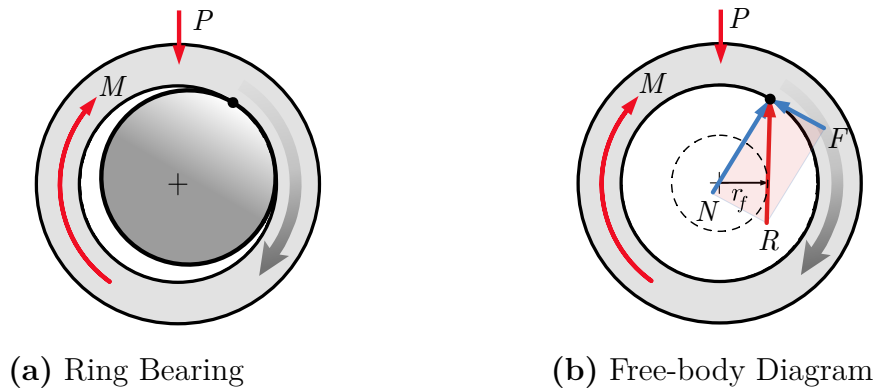
$$r_f = r \sin \phi_s = \tan^{-1} F/N.$$

4. Finally, draw a free-body diagram of the shaft with all applied loads and the friction resultant  $R$ , then solve the equations of equilibrium to find the unknowns.

### 9.6.3 Fixed Shaft and Rotating Bearing

Another type of journal bearing is designed with a fixed shaft and a rotating bearing. While the solution process is quite similar to the process covered above, the main difference is that you will draw a free-body diagram of the rotating bearing instead of the shaft.

Figure 9.6.3 shows the diagrams for a journal bearings with a fixed shaft and rotating bearing.

**Figure 9.6.3****Thinking Deeper 9.6.4 Contact Point Shift.**

In this section we saw that the contact point shifts in the direction of the relative impending motion of the bearing or opposite to the relative motion of the shaft; This is true for dry friction bearings, but with oil lubricated bearings, the shaft starts by a shift this way, but as the shaft speed increases a *hydrodynamic oil wedge* forms which shifts the shaft in the other direction in much the same way that a water skier lifts up and skims the water at high speeds.

## 9.7 Rotating Discs

**Key Questions**

- Select the appropriate disc friction equation among those for hollow circular areas, solid areas, and disc brakes with a circular arc and
- Compute the possible moment the friction forces from disc friction can resist.

### 9.7.1 Disc Friction

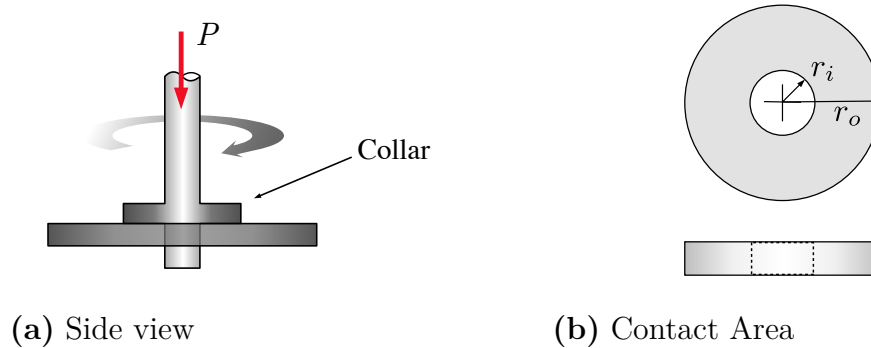
Disc friction refers friction between a flat, rotating body and a stationary surface. Disc friction exerts a moment on the bodies involved which resists the relative rotation of the bodies. Disc friction is applicable to a wide variety of designs including end bearings, collar bearings, disc brakes, and clutches.



**Figure 9.7.1** This orbital sander rotates a circular sanding disc against a stationary surface. The disc friction between the sanding disc and the surface exert a moment on both the surface and the sander.

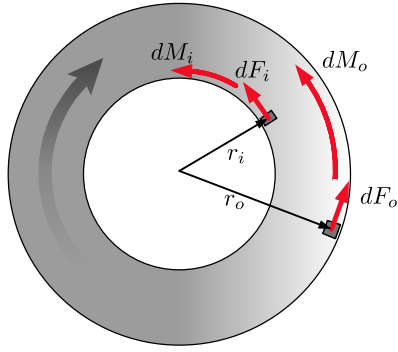
### 9.7.2 Collar Bearings

A collar bearing, shown in [Figure 9.7.2](#) has a rotating annular area in contact with a stationary bearing surface. The rotating shaft passes through a hole or bearing in the surface to maintain radial alignment. The collar bearing prevents axial motion and acts as a **thrust bearing** to transfer the axial load to a solid foundation.



**Figure 9.7.2** Collar bearing

The friction force at any point of the contact area will equal the normal force at that point times the kinetic coefficient of friction at that point. If the coefficient of friction and the pressure between the collar and the surface is the same at all points, then the same friction force at every point is the same as well. This does not mean that the moment exerted at every point is equal as well. Elements towards the outside of the contact area cause larger moments than those closer to the center since they have larger moment arms.



$$\begin{aligned} dF_f &= \mu_k dF_N \\ dF_i &= dF_o \\ r_i &< r_o \\ dM_i &< dM_o \end{aligned}$$

**Figure 9.7.3** Forces and moments on differential areas

The total moment exerted acting on the disc due to the friction forces is found by integrating the elements  $dM$  over the contact area. The moment of each element will be equal to the product of the coefficient of kinetic friction, the normal force pressure, the moment arm and the area of each element

$$dM = \mu_k p r dA.$$

$$\begin{aligned} M &= \int_A dM \\ &= \int_A \mu_k p r dA \end{aligned}$$

The coefficient of friction and pressure terms are constant so can be moved outside the integral, and since pressure is defined as force per unit area  $p = F/A$ , the pressure term can be replaced with the applied load divided by the bearing contact area,

$$p = \frac{P}{\pi(r_o^2 - r_i^2)}.$$

A differential element of area  $dA$  can be expressed in terms of radial distance  $r$  allowing us to integrate with respect to  $r$ .

$$dA = 2\pi r dr,$$

Making these substitutions leads to an equation that is easy to integrate.

$$\begin{aligned} M &= \mu_k p \int_A r dA \\ &= \mu_k \left( \frac{P}{\pi(r_o^2 - r_i^2)} \right) \int_{r_i}^{r_o} r (2\pi r) dr \\ &= \frac{2\mu_k P}{(r_o^2 - r_i^2)} \int_{r_i}^{r_o} r^2 dr \end{aligned}$$

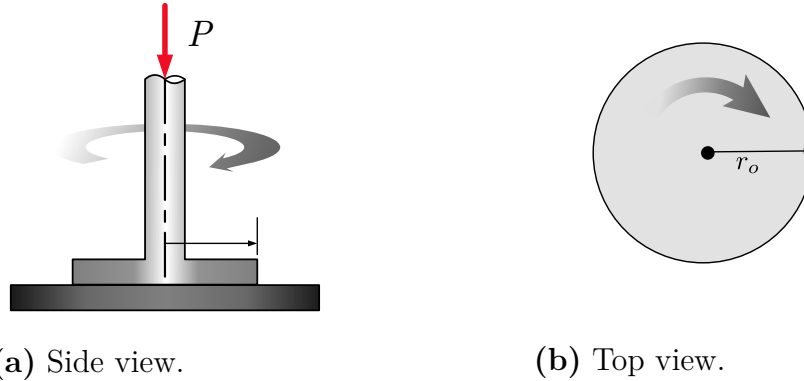
Integrating this integral, evaluating the limits and simplifying gives the final result

$$M = \frac{2}{3} \mu_k P \left( \frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} \right). \quad (9.7.1)$$

### 9.7.3 End Bearings

In cases where we have a solid circular contact area such as with a solid circular shaft, an end bearing, or the orbital sander shown in [Figure 9.7.1](#) we simply set the inner radius to zero and simplify equation (9.7.1). If we do so, the result is

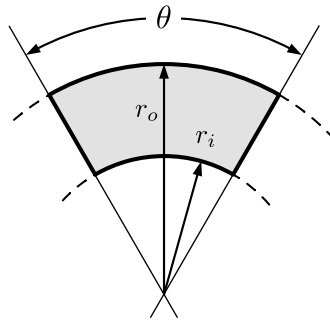
$$M = \frac{2}{3}\mu_k Fr_o. \quad (9.7.2)$$



**Figure 9.7.4** A circular thrust bearing.

### 9.7.4 Circular Arc Bearings

Automobile disc brakes have a contact area that looks like a section of the hollow circular contact area we covered earlier.



**Figure 9.7.5** The contact area in disc brakes is often approximated as a circular arc with a contact angle  $\theta$ .

Disc brakes, due to their smaller contact area, have higher pressure for the same applied force but a smaller area over which to exert friction. In the end, these factors cancel out and we end up with the same formula we found in [Subsection 9.7.2](#). Notice that this formula is independent of  $\theta$ .

Brake pad on one side:

$$M = \frac{2}{3}\mu_k P \left( \frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} \right) \quad (9.7.3)$$

Most disc brakes, however, have two pads one on each side of the rotating disc, so we will need to double the moment if so.

Brake pads on each side:

$$M = \frac{4}{3} \mu_k P \left( \frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} \right) \quad (9.7.4)$$

## 9.8 Exercises (Ch. 9)

Chapter 9 Exercises

0/140

|                                                                   |               |
|-------------------------------------------------------------------|---------------|
| <a href="#">Friction: Block on Incline</a>                        | 0/20          |
|                                                                   | Not attempted |
| <a href="#">Friction: Rope around a cylinder</a>                  | 0/10          |
|                                                                   | Not attempted |
| <a href="#">Friction: Determine direction of <math>F_f</math></a> | 0/30          |
|                                                                   | Not attempted |
| <a href="#">Friction: Range of values 1</a>                       | 0/20          |
|                                                                   | Not attempted |
| <a href="#">Friction: Range of values 2</a>                       | 0/20          |
|                                                                   | Not attempted |
| <a href="#">Friction: block on block 1</a>                        | 0/20          |
|                                                                   | Not attempted |
| <a href="#">Friction: block on block 2</a>                        | 0/20          |
|                                                                   | Not attempted |

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